RUTUJA KADAM DAY 04 – TCP/IP & OSI MODEL

OSI/TCP-IP THEORY

1. What is a network protocol?

A network protocol is a set of rules that devices follow while communicating with each other over a network. It defines how data is transmitted and received. Examples include HTTP, HTTPS, DNS, TCP, UDP, IP and Ethernet.

2. What is the OSI Model?

OSI (Open Systems Interconnection) Model is a reference model that divides network communication into seven layers. Each layer performs a specific function, making network communication easier to understand, design, troubleshoot and secure.

3. Name all seven OSI layers in correct order.

The seven OSI layers from top to bottom are:

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

Mnemonic: All People Seem To Need Data Processing.

4. What is the purpose of the Application Layer?

The Application Layer provides network services directly to user applications. It allows applications to communicate over a network using protocols such as HTTP, HTTPS, DNS, FTP and SMTP.

5. What is the purpose of the Transport Layer?

The Transport Layer provides end-to-end data delivery between devices. It uses TCP and UDP protocols and manages communication through port numbers.

6. What is the purpose of the Network Layer?

The Network Layer handles logical addressing and routing of data between different networks. It uses IP addresses (IPv4/IPv6) to identify devices and routers to forward data towards its destination.

7. What is the purpose of the Data Link Layer?

The Data Link Layer handles communication between devices on the same local network. It uses MAC addresses and Ethernet frames to transfer data. Switches operate at this layer.

8. What is the purpose of the Physical Layer?

The Physical Layer is responsible for transmitting raw bits (0s and 1s) over a physical medium. It includes network cables, fiber-optic cables, interfaces and electrical or optical signals.

9. What is the TCP/IP Model?

TCP/IP (Transmission Control Protocol/Internet Protocol) Model is a networking model used in real-world network communication. It divides communication into four layers: Application, Transport, Internet and Network Access.

10. Explain the difference between OSI and TCP/IP models.

OSI Model	**TCP/IP Model**
It has seven layers.	It has four layers.
It is a reference model.	It is used in real-world networking.
It has Application, Presentation and Session as separate layers.	These three layers are combined into the Application Layer.
It has a separate Network Layer.	It has an Internet Layer.
Data Link and Physical are separate layers.	They are combined into the Network Access Layer.

11. What is encapsulation?

Encapsulation is the process of adding headers to data as it moves down through the networking layers before transmission.

The process is:

Application Data → Transport Header → IP Header → Ethernet Header → Bits.

12. What is decapsulation?

Decapsulation is the process of removing and processing headers from received data as it moves up through the networking layers at the receiving end.

The process is:

Bits → Ethernet Frame → IP Packet → TCP/UDP Segment → Application Data.

13. Which layer uses IP addresses?

The Network Layer (Layer 3) uses IP (Internet Protocol) addresses to identify devices and route data between different networks.

14. Which layer uses MAC addresses?

The Data Link Layer (Layer 2) uses MAC (Media Access Control) addresses for communication between devices on the same local network.

15. Which layer uses TCP and UDP?

The Transport Layer (Layer 4) uses TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) for end-to-end communication between devices.

16. Which layer includes HTTP and HTTPS?

The Application Layer (Layer 7) includes HTTP (Hypertext Transfer Protocol) and HTTPS (Hypertext Transfer Protocol Secure). These protocols are used for communication between web browsers and web servers.

17. Why should cybersecurity professionals understand networking layers?

Cybersecurity professionals should understand networking layers because:

1. It helps identify how data travels across a network.
2. It helps analyse HTTP, HTTPS, DNS and API traffic.
3. It helps perform port scanning and service identification.
4. It helps analyse IP addresses and routing.
5. It helps identify MAC addresses and Ethernet communication.
6. It supports network monitoring and troubleshooting.
7. It helps detect network vulnerabilities and security threats.

This knowledge is important for using tools such as Wireshark, Nmap, Burp Suite and firewalls.

18. Explain how network data travels from a browser to a server.

When a user opens a website in a browser, data travels through different networking layers:

1. **Application Layer:** The browser uses HTTP or HTTPS to request information from the web server.
2. **Transport Layer:** TCP or UDP handles communication and uses port numbers.

3. **Network Layer:** IP addresses identify the source and destination and help route data.
4. **Data Link Layer:** MAC addresses and Ethernet frames handle local network communication.
5. **Physical Layer:** Data is transmitted as bits through network cables, fiber or wireless signals.

At the receiving server, the process of decapsulation takes place. Each layer processes and removes its corresponding header, and the original application data reaches the server.

The server processes the request and sends a response back to the browser through the same networking layers.

PRACTICAL (4 Marks)

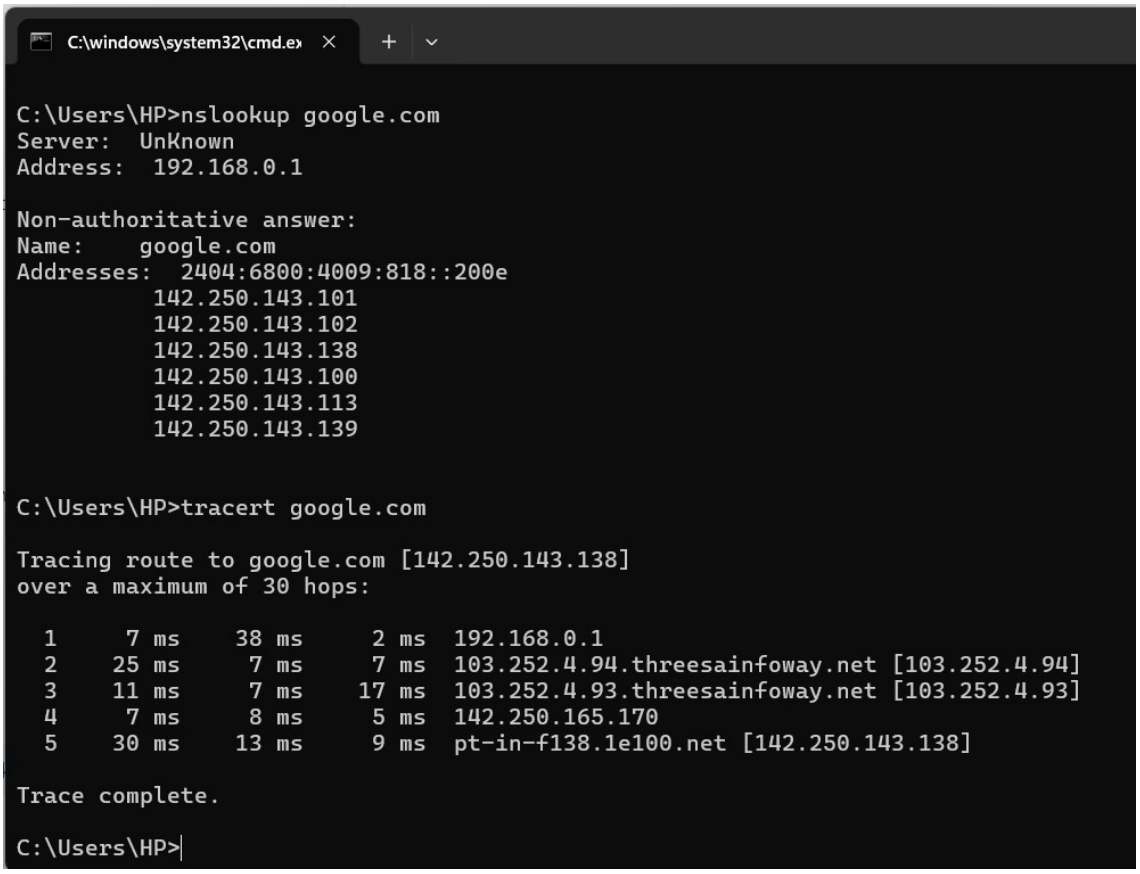
19. Your resolved IP address for google.com (nslookup).

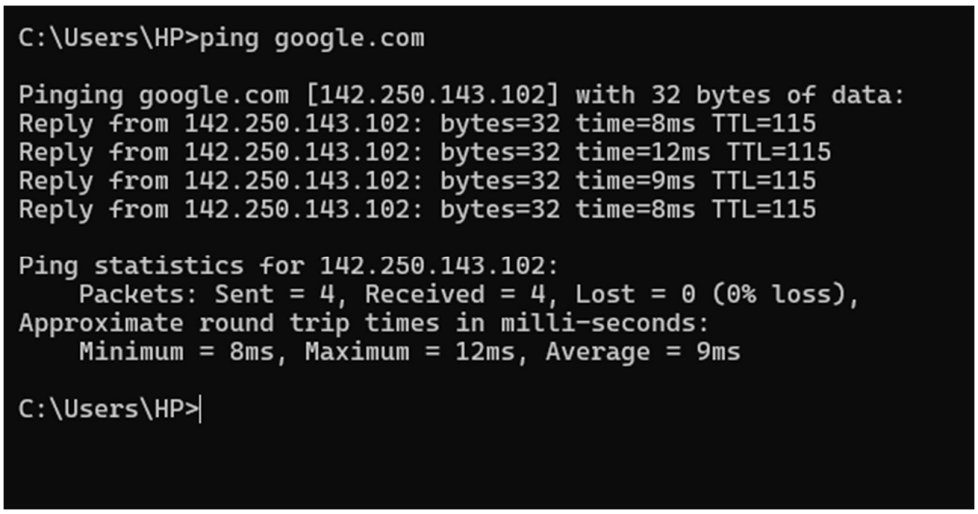
- Domain Name: google.com
- Resolved IPv4 Address: 142.250.143.138
- DNS Server: 192.168.0.1

20. Number of traceroute hops to google.com.

- Destination: google.com
- Number of Traceroute Hops: 5

The traceroute command successfully completed in 5 hops.


C:\windows\system32\cmd.ex
C:\Users\HP>nslookup google.com
Server: UnKnown
Address: 192.168.0.1
Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:818::200e
          142.250.143.101
          142.250.143.102
          142.250.143.138
          142.250.143.100
          142.250.143.113
          142.250.143.139
C:\Users\HP>tracert google.com
Tracing route to google.com [142.250.143.138]
over a maximum of 30 hops:
  1   7 ms   38 ms   2 ms 192.168.0.1
  2   25 ms   7 ms   7 ms 103.252.4.94.threesainfoway.net [103.252.4.94]
  3   11 ms   7 ms   17 ms 103.252.4.93.threesainfoway.net [103.252.4.93]
  4   7 ms   8 ms   5 ms 142.250.165.170
  5   30 ms   13 ms   9 ms pt-in-f138.1e100.net [142.250.143.138]
Trace complete.
C:\Users\HP>


C:\Users\HP>ping google.com
Pinging google.com [142.250.143.102] with 32 bytes of data:
Reply from 142.250.143.102: bytes=32 time=8ms TTL=115
Reply from 142.250.143.102: bytes=32 time=12ms TTL=115
Reply from 142.250.143.102: bytes=32 time=9ms TTL=115
Reply from 142.250.143.102: bytes=32 time=8ms TTL=115
Ping statistics for 142.250.143.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 12ms, Average = 9ms
C:\Users\HP>